

Sky-Map Anisotropy & HHT Findings — Technical Report (0V3)

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Executive Summary

- We constructed all-sky healpy Mollweide maps of the BAO-derived ratio $R_{\text{BAO}} \equiv D_{\text{H}} / D_{\text{M}}$ over $0.10 \leq z < 1.20$ using a unified catalog.
- A mask-aware, mean-centered linear dipole fit (ridge-regularized) provides a stable estimate of the large-scale anisotropy axis and effective dipole amplitude.
- Permutation tests (quick-grid runs used $n \approx 200$ per configuration) supply a null baseline for the recovered dipole amplitude; tomography bins are optionally used to test redshift dependence.
- IMF2/IMF3 Hilbert–Huang Transform (HHT) energies from prior SN residual analyses are optionally overplotted (top quantile by energy, with optional locking filter), enabling a visual check for directional coincidence with BAO features.
- We support overlay of externally supplied FR and KdS axis hypotheses in Galactic (ℓ, b), plus quick cross-power proxies via $C_1(\text{map}, \text{template})$.
- The current quick-grid sweep confirms end-to-end execution across several NSIDE/smoothing/mask/bin choices and writes PNG, FITS, and JSON summaries per run, plus a CSV aggregator.

Background (Technical, Non-Cosmology Reader)

Baryon Acoustic Oscillations (BAO) are regular, faint density ripples left in the distribution of galaxies by sound waves in the early universe.

They act as a standard ruler. From BAO measurements we infer characteristic distances as a function of redshift z .

Two such distances are the ‘radial’ scale D_{H} and the ‘transverse’ scale D_{M} . Mapping their ratio across the sky can reveal large-scale anisotropies.

- Key quantities:
- $D_{\text{H}}(z) = c / H(z)$: the radial BAO scale (‘Hubble distance’) at redshift z .
- $D_{\text{M}}(z)$: the transverse comoving distance (closely related to the line-of-sight integrated geometry).

- r_d : the comoving sound horizon at the drag epoch (sometimes appears in catalogs as a division factor).
- $R_{\text{BAO}}(z) \equiv (D_H/r_d) / (D_M/r_d) = D_H / D_M$.
- HHT/IMF context:
 - The Hilbert–Huang Transform (HHT) decomposes time/redshift-ordered residuals into intrinsic mode functions (IMFs). IMF2 and IMF3 often capture mid-band structure (‘chirp-like’ features).
 - ‘Locking’ refers to persistence/coherence of an IMF’s instantaneous frequency/amplitude relative to an external scale (e.g., BAO $1/k$), tested via phase-randomized surrogates.

Equations & Definitions

$$R_{\text{BAO}}(z) \equiv (D_H/r_d) / (D_M/r_d) = D_H / D_M$$

$$D_H(z) = c / H(z)$$

$$D_M(z) = (1+z) \cdot D_A(z) \quad (D_A \text{ is the angular-diameter distance})$$

Dipole model: $R(\hat{n}) \approx R_0 + A \cdot (\hat{n} \cdot \hat{d})$ with $R_0 = \langle R \rangle$,
 A the dipole amplitude, $\hat{d}$ the unit axis

Hemisphere contrast along $\hat{d}$: $\Delta \equiv \langle R \rangle_{\{\hat{n} \cdot \hat{d} \geq \text{med}\}} - \langle R \rangle_{\{\hat{n} \cdot \hat{d} < \text{med}\}}$;
fractional $\Delta/|R_0|$

Permutation p-value: $p \approx (\# \{A_{\text{null}} \geq A_{\text{obs}}\} + 1) / (N_{\text{perm}} + 1)$

Data & Catalog

Unified BAO catalog with columns (typical): ra_{deg} , dec_{deg} , z , $DH_{\text{over_rd}}$, $DM_{\text{over_rd}}$ (or DH , DM , rd); optional HHT fields $imf2_E$, $imf3_E$, hht_{lock} .

Coordinates are converted to Galactic (ℓ, b) as needed.

Methods

- HEALPix binning at NSIDE $\in \{32, 64\}$ with Gaussian smoothing (FWHM 6–8°).
- Mask-aware mapmaking: numerator/denominator smoothing to mitigate edge bleed; optional Galactic $|b| \geq 20^\circ$ analysis/display mask.
- Dipole estimation via centered linear regression with light ridge regularization; report amplitude A_{eff} and fractional amplitude $A_{\text{eff}}/|R_0|$.
- Null calibration via value-permutation across valid pixels; optional tomography in user-defined z-bins.
- Optional overlays: FR and KdS axis great-circles + poles; IMF2/IMF3 high-energy points (top quantile) with optional locking threshold.
- Optional cross-power proxy $C_1(\text{map}, \text{template})$ for alignment checks with supplied axis templates.

Recent Quick-Grid Runs (Structure & Outputs)

The following configurations were executed by `sm_quickgrid.py`, producing PNG, FITS, and JSON summaries per tag.

Open the corresponding `*_summary.json` to read dipole (ℓ, b), R_0 , A_{eff} , fractional amplitude, permutation p , and optional $\Delta_{\text{FR}}/\Delta_{\text{KdS}}$.

Tag	NSIDE / FWHM (deg)	Mask b (deg)	Axes	z-bins	Output (basename)
ns32_s6_nomask_auto	32 / 6.0	—	—	—	SM_ns32_s6_nomask_auto (.png/.fits/_summary.json)
ns64_s8_mask20_axes	64 / 8.0	20	FR+KdS	0.10–0.40; 0.40–0.70; 0.70–1.00; 1.00–1.20	SM_ns64_s8_mask20_axes (.png/.fits/_summary.json)
ns64_s6_mask20_bins2_fr	64 / 6.0	20	FR	0.10–0.70; 0.70–1.20	SM_ns64_s6_mask20_bins2_fr (.png/.fits/_summary.json)
ns64_s8_mask20_kds_only	64 / 8.0	20	KdS	—	SM_ns64_s8_mask20_kds_only (.png/.fits/_summary.json)
ns64_s6_mask20_bins5	64 / 6.0	20	FR+KdS	0.10–0.30; 0.30–0.50; 0.50–0.70; 0.70–0.90; 0.90–1.20	SM_ns64_s6_mask20_bins5 (.png/.fits/_summary.json)
ns32_s8_mask20_noaxes	32 / 8.0	20	—	—	SM_ns32_s8_mask20_noaxes (.png/.fits/_summary.json)

HHT IMF Findings to Date (Concise)

- IMF2 ('mid-band') shows persistent structure across SN residuals; energy ridges exhibit partial locking to BAO-like scales under certain preprocessing choices.
- IMF3 amplitude tends to track cumulative 'entropy' proxies in extended models (e.g., accretion-like terms); sensitivity tests with phase-randomized surrogates reduce spurious detections.
- Surrogate-based significance: typical grids used $O(10^2-10^3)$ surrogates; robustness improves with CEEMDAN and conservative detrending/padding settings.
- Chirp investigation: instantaneous frequency drifts (df/dz) are detectable in segments; alignment with BAO $1/k$ proxies is under active testing.

Acronyms & Variables

BAO: Baryon Acoustic Oscillations

$H(z)$: Hubble expansion rate at redshift z

D_H : Radial BAO distance (c/H)

D_M : Transverse comoving distance

r_d : Sound horizon at drag epoch

R_{BAO} : BAO distance ratio D_H/D_M

HHT: Hilbert–Huang Transform

IMF: Intrinsic Mode Function

FR: Full Relativity (hypothesized axis)

KdS: Kerr–de Sitter (hypothesized axis)

NSIDE: HEALPix resolution parameter

C_1 : $\ell=1$ cross-power coefficient

Interpreting the Sky Maps (Practical Guide)

- Check color scale limits and v-clip quantiles; ensure masked plane is excluded if comparing axes.
- Use the inset: R_0 (mean), A_{eff} (absolute dipole amplitude) and fractional % of mean, (ℓ, b) for the dipole, permutation p , and (optional) $\Delta_{\text{FR}}/\Delta_{\text{KdS}}$ angular separations.
- Tomography: compare dipole angles and amplitudes across z -bins; skipped bins indicate insufficient coverage post-mask (min-pix threshold).
- Cross-checks: compare C_1 (map, FR/KdS templates) and hemisphere contrast along the fitted axis.

Known Caveats & To-Dos

- Remove redundant duplicate '--vclip-quantiles' in sm_quickgrid command construction (cosmetic).
- Run higher N_{perm} (≥ 2000) for the most promising configurations to stabilize p -values.
- Systematics scan: vary $|b|$ mask ($15\text{--}30^\circ$), smoothing ($4\text{--}10^\circ$), and NSIDE ($32\text{--}128$) while tracking dipole axis stability.
- Tomography: ensure each bin exceeds min-pix threshold; otherwise merge adjacent bins or reduce the threshold.
- Consider inverse-variance weighting when $\sigma(D_H/r_d)$, $\sigma(D_M/r_d)$ are present; validate against unweighted maps.

Reproducibility

Quick-grid launched via sm_quickgrid.py with baseline args: --rbao-mode auto, --perm-n ~ 200 , --lock-min ~ 0.6 , --highlight-quantile ~ 0.90 , mask $|b|$ either disabled or set to 20° , and optional FR/KdS axes.